Project Smart Trigger Design Documentation for ECE/CS 3992

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***Abstract*—This document describes the design of the Smart Trigger, a modular RFID-based access-control system for the AR-15 platform. A passive 125kHz RFID tag worn by the authorized user is read by an antenna embedded in a custom 3D-printed grip. A Raspberry Pi Pico microcontroller validates the tag against an authorized list and drives a micro stepper motor that moves a blocking member out of the fire control group. The document covers the system architecture, the hardware components and their integration onto a custom PCB, the CAD modeling of the grip and actuator mounts, and the firmware processes that tie the system together. It closes with a look at how the design could grow through an off-device logging server and web application, and what that complexity would cost in power, security, and development effort.**

I. Introduction

II. System Architecture

A. Overview

The Smart Trigger is a self-contained, battery-powered embedded system that adds an electronic authorization step to the mechanical operation of an AR-15. Functionally, it is a closed loop with three stages: Sense, Decide, Act.

B. Subsystems

C. Design Requirements

D. Key Architectural Decisions

III. Hardware Components and Integration

A. Mechanical Interaction

The mechanical design has three parts:

1) a **custom grip** that replaces the factory AR-15 grip and houses the electronics, battery, and RFID antenna
2) an **actuator mount** that locates the micro stepper in the fire control pocket of the lower receiver
3) a **blocking member** driven by the motor that physically prevents the fire control group from releasing the hammer

TABLE I
Components in the Smart Trigger system

Part Number	Short Name	Short Description
Pi Pico W	Pico	Microcontroller and central brain
SAA1042	Motor Driver	Stepper motor driver
RDM6300	RFID Reader	RFID reader and translator
ANT1	Antenna	Custom wound antenna tuned to 125kHz
M1	Motor	Stepper motor to rotate shaft
R1	R	47kΩ resistor
C1	C	100 nF capacitor

To understand the innovation behind the blocking member takes some understanding of the original function of the AR-15 fire control group(trigger, hammer, sear, and safety). To fire a round the trigger must be pulled, the angle the trigger is brought to in its movement releases the hold it had on the spring loaded hammer. The freed hammer slams against onto a hook called the disconnector which holds the hammer until the trigger is back to its ready position it was in before firing the last round. The safety on the AR-15 is a physical barrier to the trigger. When is the safe position the trigger is physically blocked from movement thus preventing the hammer from moving and the overall firing of the gun. The design of our safety takes advantage of the mechanical blocker of the AR-15's native safety selector. To create the most widely applicable design we focused on locking the native safety in place until an authorized user is recognized.

B. Component Selection

C. Main Board (PCB)

1) Rationale for a Custom PCB: The project imposed a strict size constraint on the electronics enclosure: the pistol grip measures [insert grip dimensions]. A custom printed circuit board (PCB) would integrate all components listed in Table **??** into a compact form. A custom design also offered greater control over the system, since unnecessary components could be removed and only the

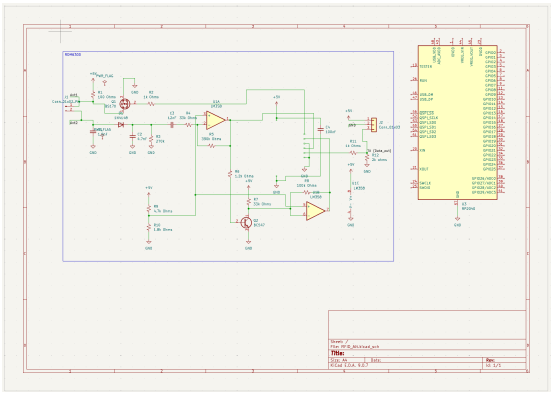


Fig. 1. First Iteration of the Custom PCB Design. The design integrates the RDM6300 RFID module, the SAA1042 stepper motor driver, and the Raspberry Pi Pico microcontroller into a single board.

required circuitry selected from each of the commercial development boards. Finally, a single integrated board would simplify installation for the end user compared with a multi-board assembly.

2) Design Process and Change in Approach: With these constraints in mind, we began by reverse engineering the RDM6300 RFID module, which serves as the interface between the Raspberry Pi Pico microcontroller and the RFID tag. We recreated the circuit in KiCad, a PCB design tool, and started to validate it through simulation in LTSpice. The resulting design is shown in Figure **??**.

This design would have allowed us to remove the RP2040 from its development board and mount it directly on the custom PCB, eliminating the many GPIO pins we did not intend to use.

Before fabricating and testing the board, however, the team questioned whether the fine-pitch leads of the RP2040 could be reliably hand soldered. Commercial manufacturers typically place such components using processes such as stencil-based solder paste application with reflow, or thermocompression bonding, neither of which was available to us. Given the project schedule and the risk of failing to mount the chip, we elected to pursue a simpler design.

3) Tradeoffs of the Revised Design: Although each of the existing boards fit within the grip, they would not be held securely in place. We therefore changed the role of the PCB to that of a perfboard-style central hub that interconnects the individual boards. This decision involved several tradeoffs. The perfboard approach retains unused features, including GPIO pins, clock circuitry, and UART pins. The boards must also be arranged to fit within the grip, making the assembly less compact than a custom PCB that integrates their functions. However, this approach required no additional parts, since all components were already in hand, and its lower complexity improved the likelihood of delivering a working product by the deadline.

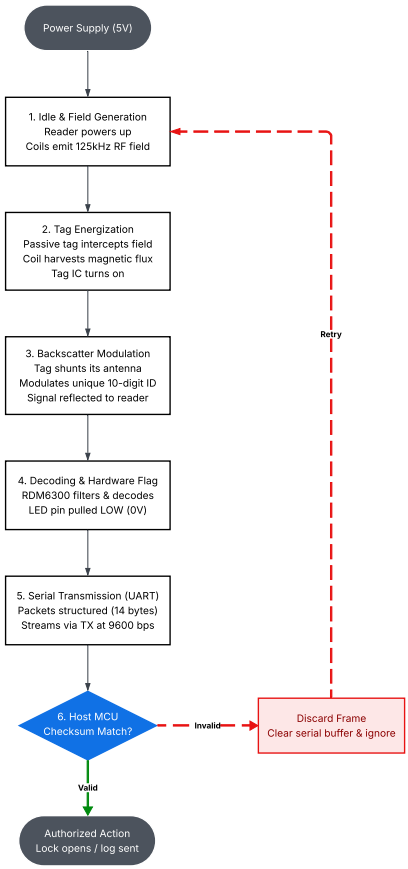


Fig. 2. A flow diagram of the RDM6300 RFID module. The module reads the RFID tag and translates its signal into a UART serial output for the Raspberry Pi Pico.

The current design is shown in Figure . The board is configured to sandwich the Raspberry Pi Pico microcontroller on one side with the motor driver on the opposite side. The RDM6300 sits directly below the motor driver and is mounted through solder points that accept the short portion of its header pins, forming a compact stack of boards.

D. RFID Board (RDM6300)
E. Micro Stepper Motor & Driver Board (SAA1042)
F. Integration and Test Plan

IV. CAD Modeling and Design

A. Scope of the Mechanical Design
B. Custom Grip
C. Actuator Mount and Blocking Member
D. Materials and Print Parameters
E. Design Iteration

V. Software Processes

A. Development Environment
B. Firmware Architecture
C. Control State Machine
D. RFID Processing
E. Actuator Control
F. Software Testing
G. Expansion: Logging Server and Web Application

VI. Conclusion

REFERENCES